

Scalable Updatable Secure Storage from Linear-Size Homomorphic Vector Commitments

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Abstract

Updatable secure storage (USS) combines updatable encryption with a homomorphic vector commitment (HVC) so that a client can outsource a dynamic encrypted database, rotate keys, and still detect a malicious server that serves a stale file. The only published USS construction (Chen–Guo–Li–Tang, ASIACRYPT 2023) instantiates HVC from the Catalano–Fiore vector commitment and therefore carries an $O(n^2)$ common reference string, which is already tens of megabytes at $n = 2^{10}$ files and infeasible at $n = 2^{14}$. We give KHVC, a KZG-based HVC with an $O(n)$ SRS that reuses a powers-of-tau setup, tight position binding from q -SDH, and statistical position hiding. We then instantiate USS over Type-3 pairings: openings live in $\mathbb{G}_1$ and are wrapped by RISE (so the server can update them homomorphically). File bytes are a separate DEM. AES-256-GCM has no update, so it does not meet a rotation policy. The DEM we measure is a random-oracle Ring-LWR key-homomorphic PRF, which is CPA updatable encryption rather than authenticated encryption: plaintext integrity stays with the stub. A Feist–Khovratovich all-opening and a precomputed vanishing-polynomial opening remove the quadratic server work of a naïve KZG FileUp/KeyUp. We implement the scheme and the 2023 baseline in Rust (BLS12-381) and measure real costs, including 1 GB files.¹ On an Apple M5 Pro, the KHVC verification SRS for $n = 1024$ is 48.3 KB and the operational parameters, including the KeyUp table, are 96.3 KB, against 50.5 MB for the 2023 Catalano–Fiore backend. A symmetric data key’s originator-usage period is under two years [9], so a ciphertext kept for years to a few decades is updated tens of times. We take 128 updates as the bound. An 8-bit noise pad on the Ring-LWR DEM guarantees 127 of them in the worst case; after 128 updates on 64 blocks the largest measured rounding error was 78, and by 255 updates the pad had overflowed. At 1 MB the DEM alone runs at 32.6 MB/s with a $2\times$ ciphertext and a 4096-byte token. End-to-end KeyUp of four 1 GB files, including the DEM scan, is 404 s and a 4240-byte token, against tens of minutes to hours if SHINE0 or RISE is applied to the same bytes. Sparse FileUp is not uniformly faster: at $n = 1024$ the Catalano–Fiore backend wins for a one-file update, and KHVC wins once about $2.6 \log_2 n$ slots change.

Keywords: vector commitment, KZG, updatable encryption, updatable secure storage, key rotation.

1 Introduction

Cloud storage is encrypted, and industry standards (PCI DSS, AWS/GCP CMEK) demand periodic key rotation. Updatable encryption (UE) [4, 5, 7, 6] lets a server re-key a ciphertext from a short token, without learning the plaintext. That is not enough for a dynamic database. After a file update, a malicious server can still return an old ciphertext that decrypts under the current key. Chen, Guo, Li, and Tang [10] introduced updatable secure storage (USS) to add

¹Artifact: <https://github.com/longchenHefei/USS-Optimized-KZG>. Reproduce with `cargo test --release -p uss-khvc`, `cargo run --release --bin fairbench`, `cargo run --release --bin rlwrbench`, and the CSVs in `experiments/results/`.

version integrity: the client keeps one short stub, FileUp and KeyUp communication does not grow with the number of files, and a cheating server is caught even if the current secret has leaked.

Putting an existing UE on the file bytes is not practical, and the reasons differ. On a 1 MB plaintext, block-wise RISE encrypts at 0.40 MB/s, a ReCrypt-style elliptic-curve KH-PRF at 1.29 MB/s, and SHINE0 at 1.93 MB/s, against 241 MB/s for AES-GCM. Nested AES matches that speed for one wrap, but decryption walks every epoch, so it cannot be the long-term cipher. AES-GCM itself has no update algorithm. The count those schemes would have to support is modest. NIST limits the originator-usage period of a symmetric data-encryption key to less than two years [9, Table 1]; PCI DSS requires a cryptoperiod and a change when it ends, and does not fix a shorter period. A file kept for about seven years is therefore updated a handful of times at that bound, about seven times under annual rotation, and about twenty times if an annual rotation is kept up for a twenty-year archive. Quarterly rotation over twenty years is about eighty updates. Monthly rotation over twenty years is about two hundred and forty, which we do not target. We take **128 updates** as the engineering bound: it covers annual rotation for more than a century and quarterly rotation for about thirty years, and it is the same knob used to configure nested UE for 128 re-encryptions [7].

The architecture that fits this bound is KEM+DEM, with the DEM itself a UE. RISE wraps openings and stays on group elements, because those updates have to be homomorphic. The file body is not AES-GCM. It is a ciphertext-independent CPA scheme: USS games that mention the body are CPA games, and OF-PTXT recomputes the file hash after decryption, so the DEM does not need a MAC. We instantiate that DEM with a random-oracle Ring-LWR key-homomorphic PRF, one negacyclic product per block. At 1 MB it encrypts and updates at 32.6 MB/s, with a $2\times$ ciphertext and a 4096-byte token that does not grow with the file. Eight bits of padding per coefficient guarantee 127 updates in the worst case, because each update adds at most 1 to the rounding error. On 64 blocks the largest measured error after 128 updates was 78, still inside the pad; after 255 updates it was already 144. So this parameter sits on the 128-update bound. It does not cover a monthly rotation of a multi-decade archive without a wider pad or a client re-encryption. End-to-end KeyUp of four 1 GB files is 404s and 4240 bytes (RISE header plus the DEM token).

The 2023 construction commits to file hashes with a homomorphic vector commitment (HVC) built from Catalano–Fiore [2]. The public parameters contain every pairwise product $g^{z_i z_j}$, i.e. $O(n^2)$ group elements. At $n = 1024$ this is already 50 MB of CRS in our implementation; at $n = 10^4$ it is several gigabytes. The 2023 scheme had no public implementation; this paper provides one, so the large-file cost is no longer a guess.

Contributions.

1. **KHVC**. A KZG [3] HVC with linear SRS, $\text{aux} = \rho \in \mathbb{F}_p$ (succinct, as required by [10]), tight q -SDH position binding (no $1/n$ loss), statistical position hiding, and public rerandomization.
2. **KEM+DEM USS**. The 2023 protocol on BLS12-381: RISE [4] on $\mathbb{G}_1$ for openings (SXDH); a CPA Ring-LWR KH-PRF as the DEM for file bytes, sized for 127 correct updates; FK20 [13] all-openings for FileUp; precomputed $\{[Z(\tau)/(\tau - \omega_j)]_1\}$ for the header half of KeyUp. KeyUp sends both the RISE token and $\Delta = s' - s$, and the server rewrites every body.
3. **Implementation and measurements**. A Rust crate with KHVC, Catalano–Fiore HVC, RISE, SHINE0, ReCrypt-style KH-PRF, nested AES-GCM, the Ring-LWR DEM, and end-to-end USS. We compare the 2023 CF-HVC backend with KHVC at the same n , and we compare payload UE, including the 128-update Ring-LWR DEM, on the same machine.

We do not claim that “hidden KZG” is new [3, 14]. The cryptographic novelty is an HVC that meets the USS interface with a linear SRS; the systems novelty is a working USS for large files, compared against the 2023 backend and against payload-only key rotation.

2 Preliminaries

Let $\mathbb{F}_p$ be the scalar field of BLS12-381 (the group order). We write $[a]_1 = ag_1$ and $[a]_2 = ag_2$. The KHVC blinding scalar is $\rho \in \mathbb{F}_p$; RISE's ephemeral scalar is $r \in \mathbb{F}_p$; a file-commitment nonce is $\nu_i \in \{0, 1\}^{256}$. These three objects are never the same symbol.

Pairings. A Type-3 pairing $e : \mathbb{G}_1 \times \mathbb{G}_2 \rightarrow \mathbb{G}_T$ on BLS12-381.

Assumption 1 (SXDH). *DDH is hard in $\mathbb{G}_1$ and in $\mathbb{G}_2$.*

Assumption 2 (q -SDH [1, 3]). *Given $([1]_1, [x]_1, \dots, [x^q]_1, [1]_2, [x]_2)$, a PPT adversary outputs $(c, [1/(x+c)]_1)$ with $c \neq -x$ only with negligible probability. The reduction uses this tuple as the KHVC SRS of degree $q = n$.*

KZG [3]. $\text{Setup}(1^\lambda, d)$ samples τ and publishes $([1]_1, \dots, [\tau^d]_1, [1]_2, [\tau]_2)$. $\text{Com}(f) = [f(\tau)]_1$. An opening of f at z is $[q(\tau)]_1$ for $q(X) = (f(X) - f(z))/(X - z)$. Evaluation binding follows from q -SDH. Commitments and openings are additively homomorphic. Verification uses only $[1]_2$ and $[\tau]_2$; it does not need $[\tau^d]_2$.

Definition 1 (HVC [10]). *A homomorphic vector commitment consists of $\text{Setup}(1^\lambda, n)$, $\text{Com}(\text{pp}, \mathbf{m}) \rightarrow (C, \text{aux})$, $\text{Open}(\text{pp}, \mathbf{m}, \text{aux}, i) \rightarrow \Lambda_i$, $\text{Ver}(\text{pp}, C, i, m, \Lambda_i)$, $\text{ComHom}(C, C')$, $\text{OpenHom}(\Lambda_i, \Lambda'_i)$. The auxiliary string aux is independent of n . Position binding: no PPT adversary outputs $C, i, m \neq m', \Lambda, \Lambda'$ that both verify. Position hiding: given C and $\{\Lambda_j\}_{j \neq i}$ as group elements, the slot- i message is statistically hidden (Theorem 2). Homomorphism: $C \oplus C'$ and $\Lambda_i \otimes \Lambda'_i$ open to the coordinate-wise sum.*

RISE [4]. ElGamal-style UE in a prime-order group: $\text{Enc}_k(M) = (rY, rG + M)$ with $Y = kG$, token $\Delta = k'/k$, update re-randomizes and multiplies C_1 by Δ . Lehmann–Tackmann prove IND-ENC and rand-IND-UPD from DDH in that group. We instantiate the group as $\mathbb{G}_1$; SXDH is the Type-3 stand-in for that DDH assumption. RISE is additively homomorphic on the message group, which is what USS needs for encrypted HVC openings. It is *not* a large-file cipher: the message is a group element.

USS [10]. A database of n files. **Store** produces a repository of ciphertexts and a stub C . **Rev** returns file i or $\perp$. **FileUp** replaces file i ; **KeyUp** rotates keys. Security games, inherited verbatim from [10]: IND-DD-UP, IND-FileUp-CPA, IND-REENC-CPA, and OF-PTXT (integrity against a cheating server who may know the secret key). Section 1 lists the hypotheses under which we instantiate those games; Section 5 gives the hops.

Remark 1 (Instantiation hypotheses). *The rest of the paper uses: (i) Type-3 BLS12-381 and SXDH; (ii) q -SDH with $q = n$ for KHVC binding; (iii) a trusted KZG SRS (knowledge of τ is forbidden to all parties after setup); (iv) SHA-256 modelled as a random oracle, both as COM and as the ring elements $a_b = H(\text{tweak} \| b)$; (v) file nonces ν_i of 256 bits of entropy; (vi) RISE in $\mathbb{G}_1$, IND-ENC and rand-IND-UPD under SXDH [4]; (vii) decisional Ring-LWR in dimension $N = 1024$, $R_q = \mathbb{Z}_q[X]/(X^N + 1)$, $q = 998244353$, rounding modulus $p = 2^{16}$, and a uniform secret in R_q . The DEM is correct for at most 127 updates: each update adds at most 1 to the rounding error, and eight noise bits absorb an error of absolute value at most 127.*

3 KHVC

Let $\Omega = \{\omega^0, \dots, \omega^{n-1}\}$ be the n -th roots of unity and $Z(X) = X^n - 1$.

Setup. Publish the degree- n KZG SRS $[1]_1, \dots, [\tau^n]_1$ ($n+1$ compressed $\mathbb{G}_1$ elements) and $[1]_2, [\tau]_2$. No $[\tau^n]_2$ is required: the pairing check below only uses $[\tau]_2 - [\omega^i]_2$. The hiding shift $[Z(\tau)]_1 = [\tau^n]_1 - [1]_1$ is free from the SRS. For KeyUp, precompute the table $W_j = [Z(\tau)/(\tau - \omega^j)]_1$ by one $\mathbb{G}_1$ FFT of the SRS (data-independent; n extra $\mathbb{G}_1$ elements, 48 KB at $n = 1024$). Table 1 reports two sizes. The verification SRS counted by `crs_bytes()` is $(n+1) \cdot 48 + 2 \cdot 96 + 48$ bytes: 48.3 KB at $n = 1024$, which is $50.5 \text{ MB} / 48.3 \text{ KB} \approx 1070\times$ smaller than Catalano–Fiore. That counter excludes $\{W_j\}$ and includes one redundant 48-byte copy of $[Z(\tau)]_1 = [\tau^n]_1 - [1]_1$. Operational parameters are the same string plus the n compressed openings W_j , hence $48.3 + 48 = 96.3$ KB. The raw ratio against the measured Catalano–Fiore CRS (50,528,400 bytes) is $50,528,400 / (49,440 + 49,152) \approx 512$.

Commit. Interpolate $f(\omega^i) = m_i$ ($\deg f < n$), sample $\rho \leftarrow \mathbb{F}_p$, set $F = f + \rho Z$ and $C = [F(\tau)]_1$. Output $(C, \text{aux} = \rho)$. The client keeps ρ ; the server never receives it. Coefficient form: F has length $n+1$, namely $(f_0 - \rho, f_1, \dots, f_{n-1}, \rho)$.

Open / Ver. $\Lambda_i = [(F(X) - m_i)/(X - \omega^i)](\tau)$; pairing check $e(C - [m]_1, [1]_2) = e(\Lambda_i, [\tau]_2 - [\omega^i]_2)$. The ρZ term does not change evaluations on Ω ($Z(\omega^i) = 0$), so no extra $\mathbb{G}_2$ SRS element appears.

Hom / ReRand. $C + C'$ and $\Lambda_i + \Lambda'_i$. $\text{ReRand}(C) = C + \rho'[Z(\tau)]_1$, which does not change any $F(\omega^i)$. If τ were known, rerandomization would be meaningless; trusted setup is essential.

Theorem 1 (Binding). *If q -SDH holds then KHVC is position binding. An adversary with advantage ε yields a q -SDH solution with advantage $\varepsilon - O(1/p)$.*

Proof. Two accepting openings of the same C at the same ω^i to $m \neq m'$ satisfy

$$e(C - [m]_1, [1]_2) = e(\Lambda, [\tau]_2 - [\omega^i]_2), \quad e(C - [m']_1, [1]_2) = e(\Lambda', [\tau]_2 - [\omega^i]_2).$$

Dividing and scaling by $(m' - m)^{-1}$ produces $[1/(\tau - \omega^i)]_1$. The reduction is given the q -SDH tuple $([1]_1, \dots, [x^n]_1, [1]_2, [x]_2)$, publishes it as the KHVC SRS with x playing the role of τ , and returns $(c, [1/(x + c)]_1)$ for $c = -\omega^i$. It does not guess i . The $O(1/p)$ loss is the probability that $m' - m = 0$ after a wrap in $\mathbb{F}_p$, which the winning condition $m \neq m'$ already excludes up to that encoding. $\square$

Theorem 2 (Statistical hiding). *Let the adversary see C and $\{\Lambda_j\}_{j \neq i}$ as $\mathbb{G}_1$ elements. Except with probability at most n/p over setup (the event $\tau \in \Omega$), the two position-hiding worlds are identical, and $\rho_0 \mapsto \rho_1$ below is a translation. An unbounded distinguisher therefore has advantage at most n/p .*

Proof. The adversary names a slot i and two vectors that agree off i . Write $\Delta m = m_{i,1} - m_{i,0}$ and let L_i be the Lagrange basis polynomial for ω^i on Ω , so the interpolated polynomials satisfy $f_1 = f_0 + \Delta m L_i$. If $\tau \notin \Omega$ then $Z(\tau) \neq 0$. Given the blinding ρ_0 of the first world, set

$$\rho_1 = \rho_0 - \Delta m \cdot \frac{L_i(\tau)}{Z(\tau)}.$$

Then

$$F_1(\tau) = f_1(\tau) + \rho_1 Z(\tau) = f_0(\tau) + \Delta m L_i(\tau) + \left(\rho_0 - \Delta m \frac{L_i(\tau)}{Z(\tau)} \right) Z(\tau) = F_0(\tau).$$

Hence C is the same group element in both worlds. For every $j \neq i$ the opening is the group element $\Lambda_j = [(F(\tau) - m_j)/(\tau - \omega^j)]_1$. It depends on F only through the scalar $F(\tau)$, and m_j is the same in both worlds, so Λ_j is the same group element. The map $\rho_0 \mapsto \rho_1$ is a translation of $\mathbb{F}_p$, so a uniform ρ_0 yields a uniform ρ_1 . The equality is numerical at the setup point τ . It is not a polynomial identity, and the adversary is not given the coefficient vector of F . $\square$

Homomorphism is the KZG homomorphism of F and of the quotients.

Lemma 1 (All-openings decomposition). *Let $F = f + \rho Z$ with $\deg f < n$ and $m_i = F(\omega^i) = f(\omega^i)$. Then*

$$\frac{F(X) - m_i}{X - \omega^i} = \frac{f(X) - m_i}{X - \omega^i} + \rho \cdot \frac{Z(X)}{X - \omega^i},$$

because $Z(\omega^i) = 0$. The first summand has degree at most $n - 2$; the second has degree $n - 1$. Equivalently, the coefficient vector of F is $(f_0 - \rho, f_1, \dots, f_{n-1}, \rho)$, and FK20 [13] is run on the length- n prefix $f - \rho$ (a degree- $< n$ polynomial). Hence $\Lambda_i = \pi_i^{\text{FK20}}(f - \rho) + \rho \cdot W_i$ with $W_i = [Z(\tau)/(\tau - \omega^i)]_1$. The table $\{W_j\}$ is computed once from the SRS by a single $\mathbb{G}_1$ FFT; query-time KeyUp does not run n MSMs. A naïve synthetic division plus MSM is $O(n^2)$ group operations; FK20 is a Toeplitz embedding, a length- $2n$ $\mathbb{G}_1$ FFT, and a length- n FFT, i.e. $O(n \log n)$ group additions. KeyUp opens the zero vector, so only the second term remains: n scalar multiplications by the precomputed W_j .

4 USS instantiation

We follow the VC-then-UE template of [10, §5]. Let $\text{COM}(m_i; \nu_i) = \text{SHA-256}(m_i \| \nu_i)$ mapped into $\mathbb{F}_p$ by interpreting the digest as a little-endian integer. In the ROM this is hiding (the nonce has 256 bits) and binding (finding $(m, \nu) \neq (m', \nu')$ with the same digest is a RO collision).

DEM (ROM Ring-LWR). The epoch key is one uniform secret $s \in R_q$, $R_q = \mathbb{Z}_q[X]/(X^N + 1)$. A file is stored under a fresh public tweak. For each block b , $a_b = H(\text{tweak} \| b)$ and

$$F_s(b) = \left\lfloor \frac{p}{q} (a_b \cdot s) \right\rfloor \in R_p.$$

The ciphertext is $\text{encode}(\nu_i \| m_i) + F_s(b)$, one ring element per 1024 payload bytes, with eight message bits and eight noise bits in each coefficient. The KeyUp token is $\Delta = s' - s \in R_q$ (4096 bytes). The server adds $F_\Delta(b)$ to every block. In $\mathbb{Z}_p$ this is a fresh encryption under s' plus a rounding error of infinity norm at most 1 per update, hence at most 127 after 127 updates, which the pad absorbs. Openings are still wrapped by RISE in $\mathbb{G}_1$. There is no per-file AES seed: body confidentiality is this DEM, not a re-wrap of a static key. We claim CPA only. OF-PTXT recomputes COM after decryption, so a mauled body that decodes to a different file fails verification, and the DEM carries no MAC.

ParGen($1^\lambda, n$). Run $\text{KHVC.Setup}(1^\lambda, n)$. Output pp .

KeyGen(pp). Sample a RISE key pair $(k, Y = kg_1)$ and a uniform DEM secret $s \in R_q$. The USS secret is (k, s) , not ρ .

Store($\text{pp}, k, s, (m_1, \dots, m_n)$). For each i : sample ν_i and a tweak; encrypt $\nu_i \| m_i$ under F_s as above; RISE-encrypt the KHVC opening Λ_i . Let $h_i = \text{COM}(m_i; \nu_i)$. The client samples a fresh ρ , sets $(C, \rho) \leftarrow \text{KHVC.Com}(h_1, \dots, h_n)$, and sends the repository (Ring-LWR bodies, tweaks, RISE openings) and stub C to the server. The client may discard ρ after Store: later FileUp/KeyUp sample a new ρ_δ or ρ_0 . The server never receives commit- ρ .

Rev($\text{pp}, k, s, \text{repo}, i$). RISE-decrypt the opening; rebuild a_b from the stored tweak; subtract $F_s(b)$; decode $\nu_i \| m_i$; recompute h_i ; KHVC.Ver the stub at i ; output m_i or $\perp$.

FileUp(pp, k , repo, i , m'_i). The client downloads file i , recovers (m_i, ν_i) , samples ν'_i and $h'_i = \text{COM}(m'_i; \nu'_i)$, and builds a sparse delta vector with a single nonzero $h'_i - h_i$. It samples ρ_δ , commits the delta, and sends the delta commitment together with a fresh Ring-LWR ciphertext of the new file under the current s and a fresh tweak. The server KHVC-opens the delta at *all* n points (FK20 if $n \geq 1024$, else the naïve all-open) and homomorphically adds each RISE-wrapped opening. Communication is $O(|m'_i|)$ plus $O(1)$ group elements, independent of n and of other files.

KeyUp(pp, (k, s) , (k', s') , repo). The client sends a RISE token $(\Delta_R = k'/k, Y' = k'g_1)$, a fresh hiding nonce $\rho_0 \in \mathbb{F}_p$, and the DEM token $\Delta = s' - s \in R_q$. The implementation returns 4240 bytes.² The server RISE-updates every opening, adds $\text{RISE.Enc}_{k'}(\rho_0 \cdot W_j)$ to each opening, adds $[\rho_0 Z(\tau)]_1$ to the stub, and adds $F_\Delta(b)$ to every body block. Bodies are rewritten. Table 8 times this full KeyUp.

Why Type-3 RISE. Openings lie in $\mathbb{G}_1$, so RISE must be instantiated there (SXDH). The 2023 paper used a symmetric pairing group, where DDH is easy; Type-3 is the sound instantiation.

Full payload update is the Ring-LWR DEM above: ciphertext-independent, CPA, and correct for 127 updates at this pad. SHINE/OCBSHINE [6] and elliptic-curve ReCrypt [5] are the same kind of DEM and are too slow on file bytes (Section 7). Theorem 3 claims body IND-CPA and body update for this DEM, not for AES-GCM.

5 Security

Theorem 3. *Under Remark 1, and writing Adv for the usual USS advantages of [10]:*

1. *KHVC is statistically position hiding (Theorem 2) and tightly q -SDH position binding.*
2. *Header and stub IND-DD-UP, IND-FileUp-CPA, and IND-REENC-CPA follow the hybrids of [10, Thms. 3–5], with RISE in $\mathbb{G}_1$ under SXDH and with every pairing check in Type-3.*
3. *OF-PTXT follows from KHVC binding and a random-oracle collision of COM (Lemma 2), even if the current RISE key and every KeyUp nonce ρ_0 are leaked.*
4. *A body is IND-CPA under decisional Ring-LWR in the random oracle, and so is a body that has been updated at most 127 times. The updated ciphertext differs from a fresh encryption under the new secret by a rounding error of infinity norm at most the number of updates. The DEM is not authenticated encryption.*

The hops below are those of [10, Thms. 3–6] under the substitutions written here. This paper does not reprove Chen–Guo–Li–Tang.

DEM in the random oracle. Let H be the random oracle used as COM and as $a_b = H(\text{tweak}||b)$. Replacing a body by uniform elements of R_p is decisional Ring-LWR for the epoch secret: each block is a rounded product $a_b \cdot s$, and a_b is a fresh oracle output. The same hop applies after KeyUp. The server is given $\Delta = s' - s$. Since s' is uniform, Δ is uniform and independent of s , and the updated block is $F_{s'}(b) + \text{encode}(\nu||m) + e$ with $\|e\|_\infty \leq t$ after t updates. For $t \leq 127$ the pad removes e on decryption, and e does not depend on the message. Distinguishing the updated body from uniform is Ring-LWR under s' . Seeing both the old and the new body does not help beyond computing F_Δ , which the server can already compute from Δ .

²Header fields: $\Delta_R \in \mathbb{F}_p$ (32 B), $Y' \in \mathbb{G}_1$ (48 B), ρ_0 (32 B), plus a 32-byte counter word (144 B). DEM token: N coefficients of $\mathbb{Z}_q$ (4096 B). Total $144 + 4096 = 4240$.

IND-DD-UP. Follow [10, Thm. 3]. H0 is the real game. H1 replaces the challenge file’s COM output by a uniform $\mathbb{F}_p$ element; the hop fails only on a random-oracle query to the challenge nonce. H2 replaces the challenge openings by RISE encryptions of uniform $\mathbb{G}_1$ elements (RISE IND-ENC under SXDH). H3 replaces the Ring-LWR body by uniform R_p elements (the DEM hop above). H4 invokes Theorem 2 on the stub and on the openings of every slot except the challenge slot. The Type-3 change is local: every pairing check is $e(C - [h]_1, [1]_2) = e(\Lambda, [\tau]_2 - [\omega]_2)$ with $\Lambda \in \mathbb{G}_1$. Nothing is extracted from $\mathbb{G}_2$.

IND-FileUp-CPA. Follow [10, Thm. 4]. The FileUp delta is one-hot. H1 hides the new COM value in the random oracle. H2 invokes RISE IND-ENC on the FK20 openings of the delta, and Ring-LWR on the fresh body under the current s (a new tweak makes a_b a fresh oracle string). H3 invokes Theorem 2 on the updated stub. The same Type-3 pairing check replaces any symmetric-group extraction step in [10].

IND-REENC-CPA. Follow [10, Thm. 5] for headers and the stub. That hop is RISE rand-IND-UPD in $\mathbb{G}_1$, together with the KHVC rerandomization $\rho_0 W_j$. Bodies are inside the claim: the DEM hop above, under the new secret s' , for at most 127 updates.

Lemma 2 (OF-PTXT under key leakage). *OF-PTXT holds by KHVC binding and COM binding in the ROM, even if the current RISE key, the DEM secret, and every KeyUp nonce ρ_0 are leaked. Updating the body is not what the integrity argument uses.*

Proof. The adversary of [10, Thm. 6] may hold the RISE key, the DEM secret, and every KeyUp nonce ρ_0 , and may drive FileUp and KeyUp. A win is a repository that verifies at slot i for a plaintext that is not the latest honest file at i .

After an honest FileUp the stub is $C' = C \oplus C_\delta$ and the honest opening at i verifies for h'_i . A stale plaintext whose commitment is $h_i \neq h'_i$, together with an opening Λ^* that verifies for C' , is two accepting openings of one slot of C' to two messages. That is a KHVC binding break, and the reduction in the binding theorem turns it into q -SDH. If the stale bytes hash to h'_i , the same transcript is a random-oracle collision for COM. The reduction does not need the RISE key.

KeyUp adds $[\rho_0 Z(\tau)]_1$ to the stub and $\rho_0 W_j$ to each opening. Both are multiples of Z , so every evaluation on Ω is unchanged. Reusing a pre-FileUp opening, which verifies the old stub for h_i , does not verify the updated stub for h_i : the updated stub opens to h'_i . A second opening to h_i is again binding. Knowledge of ρ_δ lets the adversary recompute C_δ . It does not let the adversary replace the stub the client stores. $\square$

6 Implementation

A single Rust crate (`uss-khvc`) on arkworks 0.5 (BLS12-381), `aes-gcm` (AES baselines only), and `curve25519-dalek` 4.1. File bodies in `uss.rs` are the Ring-LWR DEM; KeyUp adds F_Δ to every block. The USS timings in Table 8 are one-shot measurements from `experiments/results/uss.csv` on this build. KHVC uses `Radix2EvaluationDomain` IFFTs and an explicit $\mathbb{G}_1$ Cooley–Tukey FFT for FK20. Catalano–Fiore HVC stores $h_i \in \mathbb{G}_1$, $\hat{h}_i \in \mathbb{G}_2$, and off-diagonal $h_{i,j} \in \mathbb{G}_1$ (Type-3 verification $e(C - mh_i, \hat{h}_i) = e(\Lambda_i, g_2)$). UE baselines live in the same process so Enc/Dec/Upd numbers are comparable. The crate now selects the naïve all-open for $n < 1024$ and FK20 otherwise. Table 1 and the USS rows are one-shot timings from a CSV that predates that switch. The switch does not change the $n \geq 1024$ cells, and the small- n USS rows were not remeasured under it.

We also looked for public UE code. SHINE has a Python/C prototype [17]; ReCrypt has an abandoned Rust prototype [18]; nested AES and RLWE KH-PRF have C code [19]. IronCore `recrypt-rs` is proxy re-encryption, not UE. RISE has no public implementation; we wrote it. Primary tables use the same Rust stack; we did not mix MIRACL C timings into the main table.

7 Evaluation

Machine. Apple M5 Pro, 24 GB RAM, 15 threads, macOS 26.6.2, rustc 1.97.1, release with LTO. HVC and USS cells are one-shot timings from `experiments/results/all.csv`. Tables 2–3 are medians from `fairbench` (`experiments/results/fair_ue.csv`): a discarded warm-up, then 21 repeats at header size and 4 KB, 5 at 64 KB, 3 at 1 MB algebraic, and 7 at 1 MB AES/KH-PRF. Table 4 is the same median accounting from `rlwrbench` (`experiments/results/rlwr_ue.csv`): warm-up discarded, then 21 repeats at 4 KB, 7 at 64 KB, and 5 at 1 MB, single-threaded. We do not time RISE or SHINE0 at 100 MB (1 MB RISE Enc is already 2.6 s). A 100 MB AES/Nested figure is taken from the historical `bench` suite; ReCrypt at 100 MB is $100\times$ the 1 MB median and is labeled extrapolation.

7.1 HVC: linear vs quadratic CRS

Table 1: KHVC vs Catalano–Fiore HVC. Times in milliseconds, one-shot. The CRS column is the verification counter (`crs_bytes`): it excludes $\{W_j\}$ and includes one redundant 48-byte $[Z(\tau)]_1$. CF-HVC is omitted for $n \geq 4096$.

n	scheme	CRS	setup	Com	Open+Ver	all-open
64	KHVC	3.3 KB	17	0.86	2.20	62 (FK20) / 36 (naive)
64	CF-HVC	204 KB	298	3.86	5.22	259
256	KHVC	12.3 KB	79	1.03	2.24	326 / 216
256	CF-HVC	3.05 MB	4475	15.5	16.8	3995
1024	KHVC	48.3 KB	379	2.36	3.45	1624 / 1694
1024	CF-HVC	50.5 MB	70249	61.8	63.0	63131
4096	KHVC	192 KB	1734	6.39	7.07	7707
16384	KHVC	768 KB	8200	17.9	18.0	36495

The verification-SRS ratio at $n = 1024$ is $50.5 \text{ MB}/48.3 \text{ KB} \approx 1070$. Adding the KeyUp table gives operational parameters of 96.3 KB, and $50,528,400/(49,440 + 49,152) \approx 512$. A Catalano–Fiore CRS at $n = 2^{14}$ would be $(16384/1024)^2 \cdot 50.5 = 12.8 \text{ GB}$; ³ the KHVC verification SRS is 768 KB and the operational parameters are about 1.5 MB. Table 1 shows FK20 losing to the naïve all-open for $n < 1024$. Use the naïve path there. The published CSV predates the switch in the crate; it does not change these $n \geq 1024$ cells. KeyUp with precomputed Z -openings is 63 ms at $n = 1024$ and 1.10 s at $n = 2^{14}$, versus a full FK20 all-open of 1.62 s and 36.5 s.

7.2 Payload UE

A 48 B RISE Enc and a 4 KB AES Enc must not share a millisecond column: on the same 4 KB plaintext the costs are 2428 ns/B versus 4.21 ns/B ($577\times$). We therefore split header latency from same-size throughput. The ReCrypt-style row is a classical elliptic-curve KH-PRF on Ristretto (DDH), not the RLWE construction of [7].

Table 2: Header UE (one group element). Median milliseconds after warm-up. Token is compact and independent of file size. AES has no server token.

scheme	$ m $	Enc	Upd	Dec	$ token $	$ ct / m $
RISE ($\mathbb{G}_1$)	48 B	0.24	0.36	0.12	80 B	2
SHINE0 (Ristretto)	32 B	0.027	0.027	0.038	32 B	1

³ $(2^{14}/2^{10})^2 = 256$ and $256 \times 50.5 \text{ MB} = 12.8 \text{ GB}$.

Table 3: Same $|m|$ payload UE. Enc/Upd/Dec in MB/s. Algebraic schemes are block-wise (RISE: 48 B $\mathbb{G}_1$; SHINE0 and ReCrypt-style KH-PRF: 32 B Ristretto). Median after warm-up. KHVC-USS uses the Ring-LWR DEM of Table 4 for bodies and Table 2 for openings; the USS KeyUp comparison is Table 7.

scheme	$ m $	Enc	Upd	Dec	Enc ns/B	$ ct / m $
RISE ($\mathbb{G}_1$)	4 KB	0.41	0.27	0.76	2428	2.02
SHINE0	4 KB	1.99	1.97	1.41	503	1.00
ReCrypt-KHPRF (EC)	4 KB	1.28	1.28	7.25	782	2.00
AES-256-GCM	4 KB	237	—*	239	4.21	1.01
Nested AES-GCM	4 KB	238	231	118 [†]	4.20	1.01
RISE ($\mathbb{G}_1$)	1 MB	0.40	0.26	0.73	2490	2.00
SHINE0	1 MB	1.93	1.92	1.37	519	1.00
ReCrypt-KHPRF (EC)	1 MB	1.29	1.30	7.52	776	2.00
AES-256-GCM	1 MB	241	—*	241	4.16	1.00
Nested AES-GCM	1 MB	247	246	125 [†]	4.05	1.00

* no compact server token; a client re-encrypt is 238 MB/s at 4 KB and 240 MB/s at 1 MB. [†] decrypt after one nested update (two AES layers).

Headers are the USS workload: sub-millisecond Enc/Upd/Dec and an $O(1)$ token (Table 2). Bulk is not. On the same 1 MB plaintext, AES-GCM encrypts at 241 MB/s (4.16 ns/B). SHINE0 is 1.93 MB/s (125 $\times$ slower), the ReCrypt-style KH-PRF is 1.29 MB/s (187 $\times$), and RISE is 0.40 MB/s (600 $\times$). The 4 KB column is the same story (237 vs 1.99 vs 0.41 MB/s). AES Upd is not a token update: the server cannot refresh the body without the key. Nested AES matches Enc throughput but Dec is linear in the number of epochs. At the 128-update bound that linear walk is 128 AES decryptions, so nested AES is not a stable DEM either. A historical one-shot 100 MB AES Enc is 0.45 s; the corresponding ReCrypt figure is 81 s if the 1 MB median is scaled by 100 (extrapolation, not a timed run).

7.3 Ring-LWR DEM, sized for 128 updates

The DEM is the random-oracle Ring-LWR scheme recalled in [7, §6]: for each block index i , $F_s(i) = \lfloor (p/q)(a_i \cdot s) \rfloor$ in R_p with public $a_i = H(i)$, and the token is $\Delta = s' - s \in R_q$. It is one negacyclic product per block, not the standard-model tree PRF that multiplies once per input bit. Parameters are a performance point, not a lattice-estimator claim: $N = 1024$, $q = 998244353$, $p = 2^{16}$, eight message bits and eight noise bits per coefficient. Each update adds a rounding error of at most 1 per coefficient, so eight noise bits guarantee 127 updates. That is the 128-update bound up to a single rotation. Past that bound the pad overflows in the worst case, and a longer archive or a monthly rotation needs a wider pad or a client re-encryption. Ciphertext expansion is 2. The token is N elements of $\mathbb{Z}_q$, hence 4096 bytes, independent of $|m|$ and of the number of files. If the NTT of every a_i is cached, that table is four times the plaintext (4 MB for a 1 MB file) and is not counted in Enc/Upd/Dec; building it for 1 MB took 35 ms.

Table 4: Ring-LWR KH-PRF DEM. Median MB/s after warm-up, single-threaded reference NTT. Public a_i excluded.

$ m $	Enc	Upd	Dec	ns/B	$ ct / m $
4 KB	33.4	33.8	34.4	30.0	2
64 KB	33.0	32.8	33.7	30.3	2
1 MB	32.6	32.6	32.9	30.7	2

On 64 blocks the largest absolute rounding error was 42 after 64 updates and 78 after 128 updates, both inside the pad ($|e| < 128$). On 2 blocks it was 144 after 255 updates, which the

pad does not survive. Four 1 GB bodies at the 1 MB update median of Table 4 would be about 126 s if a_b were cached. The end-to-end USS KeyUp of Table 8 rebuilds each a_b from the public tweak and takes 404 s for the same four files, with a 4240-byte token. The server rewrites a $2\times$ ciphertext locally. This is the payload rotation the 128-update policy actually costs. It is still about an order of magnitude faster than SHINE0 or RISE on the same bytes, and AES-GCM cannot update.

Concrete cost of these Ring-LWR parameters. We estimated the DEM as unstructured LWE with the lattice-estimator [8], commit 53da598: dimension 1024, modulus $q = 998244353$, secret uniform in $\mathbb{Z}_q$, error uniform on $[-7616, 7616]$ (the centred rounding width $q/(2p)$, standard deviation ≈ 4397), and $m = 2^{20}$ samples, which is one ring element per block of a 1 GB file. Raising m to 2^{22} does not change the figure. The cheapest finite attack the estimator returned is BDD at $\approx 2^{319}$ operations; uSVP is $\approx 2^{322}$ and the dual attack is $\approx 2^{334}$. Coded-BKW did not return a finite cost on this instance. The model does not use the cyclotomic ring, so this is a generic-lattice cost of the same dimension and noise, not a reduction from ideal-SVP. It is the reason these parameters are a security point and not only a timing point: every attack the estimator finished is far above 2^{128} .

7.4 End-to-end USS

Table 3 is not a USS comparison: it times encryption engines. USS is the 2023 protocol with a new HVC backend and a measured large-file path. The two comparisons that show the construction’s advantage are Table 5 (against the only published USS) and Table 7 (against rotating the same repository without a stub).

Table 5: The ASIACRYPT 2023 USS protocol with Catalano–Fiore HVC versus this paper. HVC cells are the same-stack one-shot measurements of Table 1. Verification CRS excludes $\{W_j\}$.

	2023 CF-USS	KHVC-USS
Verification CRS, $n = 1024$	50.5 MB	48.3 KB (1070 $\times$)
Operational CRS, $n = 1024$	50.5 MB	96.3 KB (512 $\times$)
ParGen, $n = 1024$	70.2 s	0.38 s (185 $\times$)
CRS, $n = 2^{14}$	12.8 GB [§]	768 KB / ≈ 1.5 MB
KeyUp, 4×1 GB	— (no public impl.)	404 s / 4240 B

[§] $(16384/1024)^2 \cdot 50.5 \text{ MB} = 12.8 \text{ GB}$. The 512 $\times$ ratio is $50,528,400/(49,440 + 49,152)$. FileUp cost depends on k ; see Table 6.

Table 6: Server opening work for a FileUp that touches k slots, $n = 1024$. KHVC cells are measured `fileup_all_open` (FK20 does not see k). CF cells with $k < n$ are $k \cdot 63.13 \text{ s}/1024$, a linear estimate from the measured dense all-open, not a sparse CF implementation.

k	CF-HVC	KHVC	source
1	$\approx 62 \text{ ms}$	1.62 s	CF estimated, KHVC measured
8	$\approx 0.49 \text{ s}$	1.62 s	CF estimated, KHVC measured
64	$\approx 3.9 \text{ s}$	1.65 s	CF estimated, KHVC measured
n	63.1 s	1.62 s	both measured (39 $\times$ only here)

The durable gain against the 2023 backend is the CRS: at $n = 1024$ the operational parameters are 512 $\times$ smaller, and at $n = 2^{14}$ Catalano–Fiore is 12.8 GB against about 1.5 MB. FileUp is not uniformly faster. FK20 costs about 1.62 s independent of k , while the dense CF measurement scales to about $61.7k \text{ ms}$, so the crossover is $k^* \approx 26 \approx 2.6 \log_2 n$. A one-file update favours CF-HVC; the dense case $k = n$ is the only row in which KHVC is 39 $\times$ faster. That is why

Table 7: Key rotation of the *same* repository. KHVC-USS KeyUp is measured (`uss.csv`), including the Ring-LWR body scan. Baselines re-key every file body. Algebraic-file rows have no stub and no OF-PTXT.

scheme	$n \times m $	time	sent	OF-PTXT
KHVC-USS	$4 \times 1 \text{ GB}$	404 s	4240 B	yes
AES client re-enc	$4 \times 1 \text{ GB}$	17.9 s [‡]	4.0 GB	no
Nested AES Upd	$4 \times 1 \text{ GB}$	17.5 s [‡]	4.0 GB	no
ReCrypt-KHPRF (EC)	$4 \times 1 \text{ GB}$	55 min [‡]	8.0 GB	no
SHINE0 on bodies	$4 \times 1 \text{ GB}$	37 min [‡]	4.0 GB	no
RISE on bodies	$4 \times 1 \text{ GB}$	4.6 h [‡]	8.0 GB	no
KHVC-USS	$8 \times 100 \text{ MB}$	322 s	4240 B	yes
AES client re-enc	$8 \times 100 \text{ MB}$	3.5 s [‡]	800 MB	no
KHVC-USS	$64 \times 1 \text{ MB}$	8.72 s	4240 B	yes
AES client re-enc	$64 \times 1 \text{ MB}$	0.28 s	64 MB	no

[‡] Not a timed run. $T = T_{1\text{MB}} \cdot n \cdot |m| / 1 \text{ MB}$ from `fairbench` medians (AES: client re-encrypt; others: server Upd). Block-wise algebraic throughput is essentially flat in $|m|$; a 1 GB working set may still see cache and memory-bandwidth effects not present at 1 MB.

a CF-USS 1 GB experiment is missing: the 2023 CRS does not scale to the n where USS is interesting, even when its sparse FileUp is cheaper.

Against payload-only rotation of the same four 1 GB files, KHVC-USS KeyUp is measured at 404 s and 4240 B, including the Ring-LWR body scan. AES re-encrypt is 17.9 s and 4 GB on the wire; SHINE0 or RISE on the file bytes is 37 minutes to 4.6 hours. Those algebraic rows are extrapolations, not timed runs. The USS row is slower than client AES re-encrypt on this machine, but the token is $10^6 \times$ smaller than rewriting the bodies, the server does the work, and OF-PTXT is kept.

Table 8: KHVC-USS microbenchmarks from `uss.csv` (Ring-LWR bodies). FileUp wall-clock includes decrypting file i , encrypting the replacement, and the all-open of the delta; KeyUp includes rewriting every body.

n	$ m_i $	ParGen	Store	Rev	FileUp	KeyUp
8	4 KB	3.9 ms	8.0 ms	1.9 ms	4.8 ms	7.5 ms
32	4 KB	7.6 ms	34.5 ms	2.0 ms	19.5 ms	26.3 ms
64	1 MB	15.9 ms	12.3 s	0.17 s	0.40 s	8.72 s
8	100 MB	1.9 ms	155 s	12.5 s	90.3 s	322 s
4	1 GB	1.9 ms	402 s	107 s	192 s	404 s

Store, Rev, FileUp, and KeyUp all scale with total payload bytes under the Ring-LWR DEM. KeyUp of four 1 GB files is 404 s: the DEM scan dominates, and the token is 4240 bytes independent of n and of $|m|$. The extra integrity work is an FK20 of length n , not a scan of other plaintexts.

Honest costs. Our FK20 uses projective $\mathbb{G}_1$ arithmetic without a dedicated Pippenger MSM for the Toeplitz step; at $n < 1024$ it loses to the naïve loop in Table 1. The crate now switches there. The CSV behind that table is older than the switch. A production port (blst + FK20 as in Ethereum *c-kzg-4844*) would improve the $n = 2^{14}$ all-open. We did not implement a full CF-HVC USS repository at 1 GB: Table 5 compares the HVC backend that 2023 actually uses, which is the bottleneck.

8 Related work

Vector commitments [2, 11, 12] and polynomial commitments [3] are standard. Fleischhacker–Hall–Andersen–Simkin [14] use hidden KZG as a vector commitment and as a handle for extractable witness encryption. Their interface is a (hiding) polynomial commitment / VC, not an HVC: they do not provide the USS aux-independent-of- n contract, **ComHom**/**OpenHom**, or the position-hiding game with openings of the other slots as group elements. Using a polynomial commitment as a VC is common in proof systems, including Lasso [15]. Those uses likewise omit homomorphism plus short aux, which is what FileUp/KeyUp need.

Transparent FRI commitments [16] avoid a pairing SRS. A FRI opening is not a $\mathbb{G}_1$ element, so it does not plug into the RISE homomorphism used here. That is why this USS instantiation uses KZG.

UE: BLMR, RISE [4], ReCrypt [5], SHINE [6], nested AES and an RLWE KH-PRF [7]. That paper already notes that a Ring-LWR almost key-homomorphic PRF exists and moves to RLWE to widen the noise budget; with a 128-update cap the Ring-LWR pad is enough, and Section 7 measures it. USS [10] is the integrity-aware combination; this paper is its first linear-SRS instantiation. Cryptoperiods are NIST’s [9], not a new UE definition.

9 Conclusion

KHVC removes the quadratic CRS that blocked the 2023 USS at realistic n (operational parameters $512\times$ smaller at $n = 1024$). Position hiding is statistical, up to an n/p setup event. Body confidentiality and body update are Ring-LWR in the random oracle: IND-CPA for a fresh body and for a body after at most 127 updates. OF-PTXT does not need the DEM to be authenticated, and it holds if the DEM secret leaks. Sparse FileUp favours Catalano–Fiore until about $2.6 \log_2 n$ slots change. On four 1 GB files end-to-end KeyUp is 404 s with a 4240-byte token. Generic-lattice estimation of this Ring-LWR point puts the cheapest finished attack near 2^{319} , above the 2^{128} line, without using the ring.

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